National Tsing Hua University
Department of Electrical Engineering
EE429200 IC Design Laboratory, Fall 2025

Homework Assignment #3 **(6%, total 100 points)**
Logic Synthesis for ROP3 & Enigma
Assigned on Oct. 9 2025
Due by **Oct. 23 2025**

Assignment Description

In homework 1 and homework 2, you've written RTL codes for ROP3 and Enigma. Now you are going to synthesize these designs using Design Compiler and examine synthesis results.

Logic Synthesis for ROP3

1. **Synthesize rop3_smart.v and rop3_lut256.v**
   Do logic synthesis on the rop3_smart.v and rop3_lut256.v you implemented in homework 1. You can use the tcl-scripts you wrote in lab 6 to run the Design Compiler. Remember some settings such as the name of the top design and the name of the RTL file should be changed. Synthesize both your designs with clock period = 1.8ns for fair comparisons.
   - Important: Before synthesis, modify the default parameter setting to N=32 in both rop3_smart.v and rop3_lut256.v

2. **Performance comparison**
   Please compare the synthesis results of both designs, including timing, area, and power. Then, summarize your observation in performance_comparison.txt.

3. **Grading Policy (10 points)**
   Complete both 1. and 2.

Logic Synthesis for Enigma

Please perform logic synthesis for part 4 of homework 2. The area and timing of the synthesis result are used for grading, so an RTL redesign is highly encouraged. The grading is based on the following performance index:

Performance Index (PI) = Area $\times$ Timing (e.g., 1.0×10^5)
Area: Total area shown in report_area_enigma.out. E.g., 20000um^2
Timing: Clock period of the timing constraint. E.g., 5ns

For fairness, synthesis scripts are provided in **enigma/syn/**. Please use the provided script to synthesize your design, and do not modify the synthesis scripts provided except for the file list and filenames. Also, rename enigma_part4.v to enigma.v, and remember to change the module name in the .v files as well.

1. **Requirement**

   The RTLs should pass all simulations and can be synthesized. Also, the timing slack cannot be negative in the synthesis. You need to fulfill the requirements to get any points in this part.
2. **Grading Policy (90 points)**
   A. (10 points) You will receive 10 points if you **correctly fill in your PI value in the ranking list**.
   B. (80 points) You can earn up to 80 points based on your PI ranking among all eligible students. Only students who satisfy **PI** $< \mathbf{1.0 \times 10^5}$ are included in the ranking.

      The score is computed using a Gaussian-based ranking formula:

      $$Score = 80 \times (\mu + \sigma \times \Phi^{-1}\left(1 - \frac{rank - 0.5}{N}\right))\% \text{ points.}$$

      Where:
      Rank = your rank (1 for the best performer)
      N = number of eligible students in ICLAB
      $\Phi^{-1}(\cdot)$ = inverse of the standard normal cumulative distribution function
      μ and σ = normalization parameters to map ranks into score range, $\mu = 60$, $\sigma = 20$ in this homework.
   C. Additional 2% and 1% of the total score of the semester are given to the students with the lowest and second lowest PI

Deliverable

The performance_index.txt should include your area, timing and PI. Compress HW3_{student_id} into HW3_{student_id}.zip

```
HW3_{studentid}\
----------------------------------------------------------
└─rop3                               ├─enigma
  │ performance_comparison.txt       │ │ performance_index.txt
  │                                  │ │
  ├─hdl                              │ ├─hdl
  │    rop3_lut256.v                 │ │    enigma.v
  │    rop3_smart.v                  │ │    spyglass_enigma.rpt
  │    spyglass_lut256.rpt           │ │    {xxxx}.v
  │    spyglass_smart.rpt            │ ├─sim
  │                                  │ │    rtl_sim.f
  ├─report_lut256                    │ │    run_rtl_sim.sh
  │    report_area_lut256.out        │ │    test_enigma.v
  │    report_power_lut256.out       │ │
  │    report_time_lut256.out        │ └─syn
  │    syn_rop3_lut256.log           │      0_readfile.tcl
  │                                  │      report_area_enigma.out
  └─report_smart                     │      report_time_enigma.out
       report_area_smart.out         │      synthesis.tcl
       report_power_smart.out
       report_time_smart.out
       syn_rop3_smart.log
```